

# Dawson Britton

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## SUMMARY

Mechanical engineering student at Old Dominion University and IREC Recovery & Simulation Lead who designs, builds, and tests real hardware, from CAD to flight electronics. Seeking aerospace and mechanical design internships.

## EDUCATION

**Old Dominion University** Norfolk, VA  
*B.S. Mechanical Engineering* Expected 2029  
**Relevant coursework:** Statics, Materials Science with lab (SEM microstructure, grain-size analysis), Physics II

## ROCKETRY LEADERSHIP

**Recovery & Simulation Lead, IREC Team** 2025 – 2026  
*Old Dominion University Rocketry, Spaceport America Cup* Midland, TX  

- Led recovery and flight simulation for "Hailstorm," a 10,000 ft M1845 rocket (simulated 10,760 ft, Mach 0.88).
- Designed the dual-deployment recovery system: 24 in. drogue at apogee and 100 in. main at 1,500 ft.

**PROPEL Competition Lead** 2026 – Present  
*Old Dominion University Rocketry*  

- Leading the team designing and building three Level 2-class rockets for the PROPEL student rocketry competition.
- Developing simulation designs for the club's supersonic research rocket.

## PERSONAL PROJECTS

**Rocket Flight Computer** 2026  
*Arduino Nano, C++, ICM-20649 IMU, BMP280 barometer, SD logging*  

- Built and programmed a dual-deploy flight computer that detects launch, burnout, and apogee and fires recovery charges.
- Validated on hardware: 20 Hz control loop, 0.14 m/s velocity noise, backup timers within 0.2 s of spec, and inhibits that block main deployment on the ground.
- Found a sensor-library register bug by bench measurement that left the accelerometer sampling below 1 Hz; fit the flight build in 99% of the 30 KB flash.

**High-Power Rocketry Certification**  
*OpenRocket, Onshape*  

- Earned Level 1 with an Iris 3.1; building "TBD," a dual-deploy Level 2 rocket carrying my flight computer.

## WORK EXPERIENCE

**CAD Designer** May 2026 – Present  
*Signature Canvas Makers* Hampton, VA  

- Develop flat patterns in Rhino 3D for custom boat canvas from 3D scans of customer boats.
- Produce Eastman cutting-table files and build sheets that guide fabricators through assembly.

**Lead Technician** March 2025 – Present  
*Strickland Brothers 10 Minute Oil Change* Gloucester, VA  

- Diagnose technical problems, support and train technicians, and perform maintenance to spec.

**Shop Assistant** April 2021 – August 2025  
*PEB Enterprises LLC* Gloucester Point, VA  

- Created 3D models and CAD drawings for product presentations; ran trade show demos and sales.

## SKILLS & CERTIFICATIONS

**CAD:** Onshape and Rhino 3D (advanced), SolidWorks, Autodesk Inventor; 3D scan-to-pattern, CNC cut files, drawings

**Analysis & Simulation:** OpenRocket (flight and recovery simulation); currently learning ANSYS and Simulink

**Programming & Electronics:** Arduino / C++, sensor integration (I<sup>2</sup>C, SPI), embedded debugging and hardware testing

**Fabrication:** Welding, CNC, general fabrication, automotive repair

**Certifications:** Level 1 High-Power Rocketry; ASE A1 Engine Repair (May 2024)